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CSA0209 - C PROGRAMMING AMRUTHA YAZHINI A
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C03 - AT 2

FACTORIAL :-

```
#include <stdio.h>

int factorial(int n)
{
    if ((n==0) || (n==1))
        return 1;
    return n * factorial(n-1);
}

int main() {
    int n;
    scanf("%d", &n);
    printf("%d", factorial(n));
    return 0;
}
```

INPUT:- 5

OUTPUT:- 120

Explanation:-

factorial $\rightarrow$ product of integers
from $1 \rightarrow n$.

Eg:- $5 \Rightarrow 5 \times 4 \times 3 \times 2 \times 1 = 120$

Recursion $\rightarrow$ function calling itself

here, return *factorial (n-1)
in the recursive cell

$$5 \times \text{factorial}(4)$$

$$5 \times 4 \times \text{factorial}(3)$$

$$5 \times 4 \times 3 \times \text{factorial}(2)$$

$$5 \times 4 \times 3 \times 2 \times \text{factorial}(1)$$

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BINARY SEARCH

```
#include <stdio.h>
```

```
int main()
```

```
{ int a[100], n, key;
```

```
int low, high, mid, i, found = 0;
```

```
scanf("%d", &n);
```

```
for (i = 0; i < n; i++)
```

```
scanf("%d", &a[i]);
```

```
scanf("%d", &key);
```

```
low = 0;
```

```
high = n - 1;
```

```
while (low <= high)
```



```

while (low <= high) {
    mid = (low + high) / 2;
    if (a[mid] == key) {
        printf ("%d", mid);
        found = 1;
        break;
    }
    else if (a[mid] < key) {
        low = mid + 1;
    }
    else {
        high = mid - 1;
    }
}
if (found == 0) {
    printf ("-1");
}
return 0;
}

```

INPUT:-

5

10 20 30 40 50

30

OUTPUT:-

2